

AJCR Horizon Status and Route-Divergence Brief

Human mathematician handoff

24 August 2026

Snapshot: 2026-08-24 10:06 +0800. **Source ledger:** a58a86aca1.

Purpose and verdict

This is a status handoff against the two supplied review PDFs. It records what Horizon has completed relative to the reviewed route, what remains, and why repeated sessions have not crossed the same Phase 7 boundary.

The rank-one route has reached the separably-closed milestone:

```
PicRankOneOpen / DivRankOneOpen
-> canonicalRankOneAbelIso
-> translated rank-one cover over a separably closed field
-> pic0_sepClosed_representableBy
```

The current blocker is finite-Galois descent, not the old family-level rank-one producer. Horizon is pursuing:

```
finite-stage atlas and GlueData
-> P.gluedOver and glued base-change maps
-> universal class and overlap/glue equivalences
-> binder-free RepresentableBy P.gluedOver
-> exact-carrier FiniteInAffine / projectivity
-> original-field pic0_representableBy
-> PicRepDatum -> JacobianData -> Challenge.Jacobian
```

The first line is infrastructure. The remaining lines contain three missing mathematical producers and a resource-bound compile frontier.

Reviewed route versus current status

The binding route in the review plan is:

```
PicRankOneOpen -> rankOneAbelIso
-> rankOne_translate_cover_sepClosed
-> pic0_sepClosed_representableBy
-> finiteGaloisDescent -> pic0_representableBy -> JacobianData
```

Phase	Board	Measured meaning
P3 rank-one loci	Done	Public <code>PicRankOneOpen/DivRankOneOpen</code> , openness, and base change are reported unconditional.
P4 rank-one Abel iso	Done	<code>canonicalRankOneAbelIso</code> is root-reachable and consumed by the separably-closed route; this is the family-level producer emphasized by the reviewer.
P5 translated cover	Done	The translator is tied to the input Picard class and lands in the public rank-one locus.
P6 sep-closed Pic0	Done	<code>pic0_sepClosed_representableBy</code> and same-carrier <code>picRepDatumSepClosed/jacobianDataSepClosed</code> are present.

P7 finite-Galois descent	Blocked	No unconditional original-field producer. The finite-stage glue cone is only partly native-built; universal Yoneda and exact-carrier orbit gates remain open.
P8 Jacobian	Blocked	No original-field <code>J/rep</code> pair reaches <code>PicRepDatum</code> , <code>JacobianData</code> , and <code>Challenge.Jacobian</code> .

Remaining gates

Gate	Kind	Closure condition
Native glue root	Build	Fresh root import certifies <code>GluingsDiagramIso</code> , top <code>PreSnd</code> , <code>OverlapIsoSnd</code> , and <code>GluedComparison</code> . <code>GluePackage.olean</code> appeared on 24 Aug 00:29; these four downstream artifacts remain absent.
Finite-stage representation	Math	A theorem of the form <code>pic0RepresentableBy_finiteStageGlue</code> returns <code>RepresentableBy P.gluedOver</code> , with a natural <code>homEquiv</code> and no <code>(rep : ...)</code> binder.
Exact-carrier orbit geometry	Math	Prove <code>P.gluedMap.IsProjective</code> , <code>FiniteInAffine P.glueData.glued</code> , or an equivalent result for the same carrier. Existing orbit lemmas consume this hypothesis.
Original-field <code>Pic0</code>	Math	A descent theorem returns both the scheme and its <code>RepresentableBy</code> certificate, immediately consumed by <code>pic0_representableBy</code> .
Jacobian handoff	Integration	Feed the same original-field <code>J/rep</code> through <code>PicRepDatum</code> and <code>JacobianData</code> , then into <code>Challenge.lean</code> without an import cycle.

Why the same issues recur

1. **The post-P6 route is over-decomposed.** Horizon split one producer-level Phase 7 edge into transition models, triple overlaps, ring maps, glue records, base-change isomorphisms, overlap projections, and only later the Yoneda equivalence. Prerequisites create commits but do not close an acceptance edge.
2. **Source labels are over-read.** The graph marks some declarations `lean_ok` even when their native artifacts are absent. `Pic0CriticalPath.lean` imports the top glue modules, so only a native artifact plus fresh root import and axiom print is evidence.
3. **The compile cone consumes the iteration budget.** The `GluePackage` probe used about 7.1 GB RSS. The first expensive declaration in the next cone is `overlapBaseChangeIso_hom_iota` at `Pic0FiniteStageGluingsDiagramIso.lean:276`; historical checks reached about 610 seconds.
4. **Orchestration leaves dead work looking active.** Run 0154 metadata still says `running` from 2026-08-23T14:55:00Z, while `horizon ps` reports a zombie marker and no live Lean process. Its system sessions returned the task to `queued`. Runs 0145 and 0148 repeatedly relaunched the same alignment task; later sessions mostly returned `exit 1` or `queued`.
5. **The custom carrier is an architectural question.** The review plan asks to reuse the AJC finite-Galois quotient/gluings engine. The current route first builds `P.gluedOver` and then asks for a new universal `homEquiv`. This may be right, but it may also be a second descent stack without a proved identification with the required `Pic0` functor.

Run chronology

Run	Date	Result
0145	14 Aug	Eight-round alignment task; later sessions repeatedly returned <code>exit 1/queued</code> .
0148	14 Aug	Same alignment task relaunched; metadata/audit work, then queued or failed sessions.
0149	14 Aug onward	Audit classified 266 commits as 0 acceptance edges, 82 consumed prerequisites, and 10 conditional consumers.
0153	22 Aug	Compile isolation measured the <code>GluePackage/DiagramIso</code> frontier.
0154	23–24 Aug	Compile repair produced <code>GluePackage.olean</code> , but no root-certified downstream top cone; stale <code>running</code> metadata remains.
0156	23 Aug	Roadmap decomposition separated build, Yoneda, orbit, and original-field gates; no theorem edge was added.

Questions for review

1. Does `P.gluedOver` implement the finite-level representer required by the review plan, including its universal class and natural Yoneda equivalence, or is it only an atlas/base-change model?
2. Can the existing `AJC representableBy.of.finiteGalois.baseChange` theorem consume a smaller finite-level producer instead of extending this custom glue cone?
3. What mathematical input proves orbit affineness or projectivity for the exact glued carrier?
4. Is the missing binder-free `RepresentableBy` theorem a functor/sheaf descent argument rather than another overlap identity?
5. Which single theorem should be the next bounded acceptance milestone?

Reading and evidence

MainProjects/Algebraic-Jacobian-Challenge/informal/Lean_Algebraic_Jacobian_Complete_Execution_Plan.pdf
 MainProjects/Algebraic-Jacobian-Challenge/informal/AJCR_Runs_121_122_123_Supervision_Note_2026-08-07.pdf
 MainProjects/Algebraic-Jacobian-Challenge-Rebuild/AlgebraicJacobian/Picard/Pic0CriticalPath.lean
 .archon-horizon/runs/0152/sessions/0006-horizon-ajcr-strategy-review/report.md
 .archon-horizon/runs/0153/sessions/0002-horizon-ajcr-compile-isolation/report.md
 .archon-horizon/runs/0154/sessions/0002-horizon-ajcr-compile-frontier-repair/report.md
 .archon-horizon/runs/0154/sessions/0005-horizon-ajcr-compile-frontier-repair/report.md
 .archon-horizon/runs/0156/sessions/0002-horizon-ajcr-roadmap-decomposition/report.md

The existing rebuild README is useful but its artifact inventory predates the latest `GluePackage.olean`; do not use that inventory as current. The task database currently has 28 done, 3 blocked, 4 queued, 1 running, 1 failed, and 45 cancelled tasks. The dashboard also has a blank `generatedAt` and stale Aug 21 status PDFs.

Bottom line. Horizon is not stuck because the rank-one route failed. It is stuck because Phase 7 interleaves a resource-bound custom glue cone with missing mathematical producers. A human decision on whether `P.gluedOver` is the right finite-level object, followed by one producer-level acceptance test, is needed before more sessions can be expected to move the headline theorem.